

Credential rotation

Rotating credentials in a high-availability database deployment with the objective to ensure zero downtime can be a challenge.

Here are some explicit tasks which are required to accomplish the changing of a password for an important database role like `gitlab-superuser`.

Change issue creation

Create a production change issue to track this work:

```
https://gitlab.com/gitlab-com/gl-infra/production/-/issues/new?issuable_template=change_management
```

Label the issue for criticality level 2 (C2) and severity level 2 (S2) so that production deployments are not initiated during the procedure.

Make a comment in the issue with the following content to apply some required labels:

```
/label ~change ~C2 ~S2 ~Database ~"Service::Postgres" ~"Service::Patroni" ~"requires comment"
```

If the operation is to commence immediately, use the `~"change::in-progress"` label instead of `~"change::scheduled"`. This enables deployment automation to recognize the change issue as a blocker.

Operator workstation setup

In order to support commands like `bundle exec knife <action>` it is expected that an operator will change directory to their local workstation clone of the `gitlab-com/runbooks` project and install the required ruby dependencies:

```
rbenv install
ruby -S gem install bundler
bundle install --path=vendor/bundle
```

Procedure

Phase one

- ☐ Specify the environment in which to conduct operations:

```
export GITLAB_ENVIRONMENT='gstg'
```

- ☐ Specify the link to this issue:

```
export issue_link='https://gitlab.com/gitlab-com/gl-infra/reliability/-/issues/10961' #
```

- ☐ Specify the username whose password you wish to rotate:

```
export GITLAB_USERNAME='gitlab-superuser' # CHANGE ME (as necessary)
```

- ❑ Copy the current user password:

```
bin/gkms-vault-cat gitlab-patroni "${GITLAB_ENVIRONMENT}" | jq --raw-output '"gitlab-p
```

- ❑ Record the password in a field of type Password in a secure note entitled “gitlab-patroni \${GITLAB_ENVIRONMENT} \${GITLAB_USERNAME}” in 1Password for reference in case a roll-back is necessary.

- ❑ Deploy the scripts from https://gitlab.com/gitlab-com/runbooks/-/merge_requests/2197 on each patroni node:

```
export mode=0700 install_dir='/root/scripts' repository='https://gitlab.com/gitlab-com/
for patroni_node in $(bundle exec knife search node "fqdn:patroni-*-db-${GITLAB_ENVIRON
```

- ❑ Create a new password:

```
export new_password=$(openssl rand -base64 4096 | tr -dc A-Za-z0-9 | head -c64)
echo "${new_password}" | pbcopy
echo "export NEW_PASSWORD=${new_password}" | tee ./new_password.sh &>/dev/null
```

- ❑ Record the new password in a field of type Password named “Temporary PostgreSQL superuser role password” in the environmentally appropriate “Postgres <username>” Password entry in 1Password.

- ❑ Select the first member node of the patroni cluster:

```
export patroni_node=$(bundle exec knife search node "fqdn:patroni-*-db-${GITLAB_ENVIRON
echo "${patroni_node}"
```

- ❑ Ask the first patroni node to identify the leader patroni node:

```
export leader_patroni_node=$(ssh "${patroni_node}" 'test -e /usr/bin/jq && sudo /usr/lo
echo "${leader_patroni_node}"
```

- ❑ Copy the password to the patroni leader:

```
scp ./new_password.sh "${leader_patroni_node}":/tmp/new_password.sh
bundle exec knife ssh "fqdn:${leader_patroni_node}" "sudo mv /tmp/new_password.sh /root
```

- ❑ Dry-run the script to create a new temporary database user role on the leader and record the output:

```
bundle exec knife ssh "fqdn:${leader_patroni_node}" 'sudo /root/scripts/user-role-creat
```

- ❑ Confirm that there were no relevant errors in the dry-run invocation.

- ❑ Run the script to create a new temporary database user role on the patroni leader and record the output:

```
bundle exec knife ssh "fqdn:${leader_patroni_node}" 'sudo /root/scripts/user-role-creat
```

- ❑ Confirm that there were no relevant errors in the wet-run invocation.

- ❑ Record the verbatim character string of the new user role in a field of type Text named “Temporary PostgreSQL superuser role username”

in the environmentally appropriate “Postgres <username>” Password entry in 1Password.

- Wait for replication to “catch up” to the changes in the database of the leader.

- Optionally check each node in the patroni cluster to confirm that the new temporary user role exists in each database:

```
bundle exec knife ssh --concurrency 1 "fqdn:patroni-*-db-${GITLAB_ENVIRONMENT}*" 's
```

- Create (but DO NOT yet merge) a chef MR to change the username defined in `patroni.yml` for the `GITLAB_USERNAME` user role to the name of the new temporary user in the `gitlab-cookbooks/chef-repo/roles/${GITLAB_ENVIRONMENT}-base-db` file, by committing changes to:

- Set the `default_attributes.gitlab-patroni.patroni.users.superuser.username` field to the name of the new temporary user, and also...
 - Set the `default_attributes.gitlab_walg.backup_user` field to the name of the new temporary user.

- Add a link to the MR here: For example: Configure the staging patroni fleet to use a temporary role with a time-stamped username

- Block/disable the chef-client service on all patroni hosts with an explanation that includes a link to the issue created to track this work:

```
read -p "Operating in environment ${GITLAB_ENVIRONMENT}; press return to continue, CTRL
```

- Confirm that the chef-client service has been stopped on all the patroni nodes:

```
bundle exec knife ssh --concurrency 1 "fqdn:patroni-*-db-${GITLAB_ENVIRONMENT}*" 'sudo
```

- Update the password in the GKMS vault at `gitlab-patroni.patroni.users.superuser.password` to be the temporary database role password (created above) instead of the original password for the original `GITLAB_USERNAME` user role:

```
EDITOR=`which vim` bin/gkms-vault-edit gitlab-patroni "${GITLAB_ENVIRONMENT}"
```

- Notify relevant parties about a configuration change to the `${GITLAB_ENVIRONMENT}` patroni fleet.

- Merge the MR and apply the changes if necessary. This will not actually apply the changes.

- Undo the disablement of the chef-client service on all the patroni nodes at once:

```
bundle exec knife ssh --concurrency 1 "fqdn:patroni-*-db-${GITLAB_ENVIRONMENT}*" 'sudo
```

- Invoke chef-client on all the patroni nodes in order to apply the changes:

```
bundle exec knife ssh --concurrency 1 "fqdn:patroni-*-db-${GITLAB_ENVIRONMENT}*" 'sudo
```

- Verify and record in a comment that WAL-G replication push operations are still running successfully:

```
bundle exec knife ssh "fqdn:${leader_patroni_node}" 'sudo tail --lines=6 /var/log/wal-g/wal-g.log'
```

- Verify and record in a comment that WAL-G backup write operations are still running successfully:

```
for patroni_node in $(bundle exec knife search node "fqdn:patroni-*-db-${GITLAB_ENVIRONMENT}"); do
```

- Delete the temporary new password file:

```
bundle exec knife ssh "fqdn:${leader_patroni_node}" 'sudo rm /root/scripts/.new_password'
```

Phase two

Now that the original superuser role is not being used by the patroni cluster or the replication processes, update the password for the original superuser role, and revert the configurations to use the original role.

- Ask the first patroni node to identify the leader patroni node:

```
export leader_patroni_node=$(ssh "${patroni_node}" 'test -e /usr/bin/jq && sudo /usr/local/bin/jq -r ".roles[0].name"' | head -n 1)
echo "${leader_patroni_node}"
```

- Dry-run the script to update the original GITLAB_USERNAME role with the new password on the patroni leader and record the output:

```
bundle exec knife ssh "fqdn:${leader_patroni_node}" 'sudo /root/scripts/user-role-password-update --dry-run'
```

- Confirm that there were no relevant errors in the **dry-run** invocation.

- Run the script to set the password of the original GITLAB_USERNAME role in the database to the new password on the patroni leader and record the output:

```
bundle exec knife ssh "fqdn:${leader_patroni_node}" 'sudo /root/scripts/user-role-password-update'
```

- Confirm that there were no relevant errors in the **wet-run** invocation.

- Wait for replication to “catch up” to the changes in the database of the leader.

- Optionally confirm that the change has replicated to each patroni node (You will be repeatedly prompted to enter the new password, so it is recommended that you **turn off any screen-sharing or recording**. If you paste the new password correctly, but the credentials update has not yet been replicated to all nodes in the patroni cluster, then this error will be displayed: **psql: FATAL: password authentication failed for user "\$GITLAB_USERNAME"**):

```
for patroni_node in $(bundle exec knife search node "fqdn:patroni-*-db-${GITLAB_ENVIRONMENT}"); do
```

- Create (but DO NOT yet merge) a chef MR to change the username defined in `patroni.yml` for the `GITLAB_USERNAME` user role from the name of the temporary user back to the name of the original user in the `gitlab-cookbooks/chef-repo/roles/${GITLAB_ENVIRONMENT}-base-db-patroni.json` file, by committing changes to:
 - Set the `default_attributes.gitlab-patroni.patroni.users.superuser.username` field back to the name of the original `GITLAB_USERNAME` user role, and also...
 - Set the `default_attributes.gitlab_wale.backup_user` field back to the name of the original `GITLAB_USERNAME` user role, and also...
 - Set the `default_attributes.gitlab_walg.backup_user` field back to the name of the original `GITLAB_USERNAME` user role.

- Add a link to the MR here: For example: Configure the staging patroni fleet to use the original superuser role

- Block/disable the chef-client service with an explanation on all patroni hosts:

```
read -p "Operating in environment ${GITLAB_ENVIRONMENT}; press return to continue, CTRL"
```

- Confirm that the chef-client service has been stopped on all the patroni nodes:

```
bundle exec knife ssh --concurrency 1 "fqdn:patroni-*-db-${GITLAB_ENVIRONMENT}*" 'sudo
```

- Notify relevant parties about a configuration change to the `${GITLAB_ENVIRONMENT}` patroni fleet.

- Merge the MR and apply the changes if necessary.

- Undo the disablement of the chef-client service on all the patroni nodes at once:

```
bundle exec knife ssh --concurrency 1 "fqdn:patroni-*-db-${GITLAB_ENVIRONMENT}*" 'sudo
```

- Invoke chef-client on all the patroni nodes in order to apply the changes:

```
bundle exec knife ssh --concurrency 1 "fqdn:patroni-*-db-${GITLAB_ENVIRONMENT}*" 'sudo
```

- Verify and record in a comment that WAL-G replication push operations are still running successfully:

```
bundle exec knife ssh "fqdn:${leader_patroni_node}" 'sudo tail --lines=6 /var/log/wal-g
```

- Verify and record in a comment that WAL-G backup write operations are still running successfully:

```
for patroni_node in $(bundle exec knife search node "fqdn:patroni-*-db-${GITLAB_ENVIRON
```

Roll-back

- ☐ In order to undo these changes it is recommended that the procedure be repeated with the old credentials exchanged for the new credentials.

TODO

- ☐ Delete the temporary superuser role which is no longer being used by any patroni node or database operation.